

Minos: the decision value of a calibrated uncertainty

A decision–calibration gap and a label-free validity floor for quantitative MRI

(author list)

Draft — speculative build

Provisional build. The theory (§4–5) is self-contained and machine-checked. The *applied* sections (§6–7) consume **Fashion**’s calibrated IVIM posteriors and are contextualized by **Gauge**’s coverage, identifiability wall, and monitor. Both papers are in review; this draft assumes they land as submitted. Every assumption-dependent number is marked ^P and is regenerated from seeded results by `consistency.py`. If Fashion’s ruler or Gauge’s results change in revision, the ^P numbers are invalidated and must be re-run.

Abstract

A reported uncertainty is usually judged by *coverage*: does the stated interval contain the truth at its nominal rate? But an uncertainty estimate is used to *make a decision*, and we ask a different question — what is the error bar *worth*, and when should it be *distrusted*? We give two analytic answers. First, statistical and decision calibration are different objectives that diverge under a misspecified (skewed) posterior: the decision-optimal error-bar scale departs from the coverage-optimal one by a *decision–calibration gap* $G = (1/6) |z^*(\lambda)| \gamma$ to leading order in the posterior skewness γ , with a coefficient that is a pure decision-boundary quantity (Theorem 1). Second, a label-free monitor that watches for a stale calibration recovers the *observable* component of deployment drift (bounded-detectable, $AUC \rightarrow 1$) but is provably blind to a *hidden* component that induces the same regret ($AUC = 1/2$, Theorem 2) — a principled limit, and the case for labeled spot-checks. We then apply both to intravoxel incoherent motion (IVIM) diffusion MRI. On Fashion-calibrated posteriors^P the decision–calibration gap is real and clinically material, though dominated by the reported bar’s own miscalibration rather than the small-skew law; the label-free monitor^P fires on observable shifts (AUC up to 1.00) and is blind to the high- D^* identifiability wall ($AUC = 0.52$), exactly the observable/hidden split.

I Introduction

Quantitative imaging increasingly reports a per-voxel uncertainty alongside each parameter estimate. The dominant question asked of that uncertainty is *statistical calibration*: a nominal- L interval should contain the truth a fraction L of the time. Calibration is necessary, but it is not the question a *user* asks. An error bar is consumed by a decision — treat, spare, escalate — and a decision has an asymmetric cost. This paper asks two questions about the *decision value* of a reported uncertainty:

1. **When is calibrating for coverage the wrong objective for the decision?** We show the two objectives diverge under posterior skew, by a gap we compute in closed form.
2. **When should a deployed error bar be distrusted, using no labels?** We show label-free monitoring has a hard floor: it cannot see a hidden channel of drift that nonetheless changes the decision.

Both answers are first proved in an idealized model and then applied to IVIM, where the perfusion parameters (D^* , f) are notoriously hard to estimate and where calibrated uncertainty is the object of active study.

2 Positioning

Minos does not propose a new calibration method; it prices and bounds existing ones. The relevant *baselines* are: **loss-calibrated Bayesian inference** [1, 2], which tunes a posterior so the *decision* is optimal; **decision calibration** [3], which asks a predictor’s probabilities to be calibrated for a class of decisions; and **decision curve analysis** (DCA) [4], which prices a single threshold’s net benefit. Each targets one object. Minos instead computes *how far* the decision-optimal scale departs from the coverage-optimal one (the gap scaling law — a quantity these lines do not produce, since each targets a single calibrated object), and proves the *detectability limit* of monitoring such a correction for staleness. The OOD / distribution-shift-under-uncertainty literature [5] studies calibration under shift empirically; Theorem 2 adds the label-free detectability floor it does not state.

3 The model

We use the posterior-centric model locked against the Minos-Core simulation code. The report centre is μ and the truth is $\theta = \mu + s u$, where u is a zero-mean, unit-variance skew-normal error of standardized third cumulant γ ; to leading order the true per-report posterior is the Gram–Charlier expansion $p(\theta | \mu) = \frac{1}{s} \varphi(\frac{\theta - \mu}{s}) [1 + \frac{\gamma}{6} \text{He}_3(\frac{\theta - \mu}{s})]$. The reported posterior is the Gaussian $\mathcal{N}(\mu, (\tau s)^2)$; τ is the calibration knob. The piecewise-linear utility has a single active threshold t_2 (treat/escalate), with under:over cost asymmetry $\lambda = k_{\text{under}}/k_{\text{over}} \geq 1$. We write τ_{stat} for the coverage-optimal scale and τ^* for the decision-optimal scale; the gap is $G = \tau^* - \tau_{\text{stat}}$.

Assumption 1 (regime). Expansions are leading-order in γ (neglected terms $O(\gamma^{4/3})$, $O(\gamma^2)$); a single active threshold; bounded Lipschitz utility ($\text{Lip}_\theta U = k_{\text{under}} = 3.0$); τ^* identified only where the expected-utility curve is non-flat.

4 Theorem 1 — the decision–calibration gap

Theorem 1 (gap scaling law). *Under the locked model, to leading order in the posterior skewness γ ,*

$$\tau_{\text{stat}}(\gamma) = 1 - a\gamma + O(\gamma^2), \quad a = 0; \quad \tau^*(\gamma) = 1 + \frac{1}{6} |z^*(\lambda)| \gamma + O(\gamma^{4/3});$$

$$G(\gamma) = \tau^* - \tau_{\text{stat}} = \frac{1}{6} |z^*(\lambda)| \gamma + O(\gamma^{4/3}),$$

where $z^*(\lambda) < 0$ solves $(\lambda - 1)\psi(z) + z = 0$ with $\psi(z) = z\Phi(z) + \varphi(z)$.

The coverage scale is *first-order skew-insensitive* ($a = 0$, the odd Gram–Charlier correction integrates to zero over a symmetric interval), so the entire first-order gap is decision-side; the coefficient is purely the decision-boundary offset $z^*(\lambda)$ and does not depend on where the population sits relative to the threshold. All coefficients are sympy-derived (theory GATE 1). At the default cell ($\lambda = 3$, $\gamma = 0.667$) the leading-order $\tau^* = 1.0485$ and the full gap $G = 0.0847$ ($z^*(3) = -0.436$, slope $|z^*(3)|/6 = 0.0727$); the theory reproduces the published v2/v3 sweeps to within tolerance with no fitted constant (GATE 3).

5 Theorem 2 — the label-free detectability floor

Theorem 2 (detectability floor). *Split deployment staleness into an observable channel (the reported point law shifts) and a hidden channel (only the truth shifts). The observable stale-correction regret is bounded-detectable, $R_{\text{obs}} \leq L_U \cdot W_1(P_O^{\text{dep}}, P_O^{\text{cal}})$ with $L_U = k_{\text{under}} = 3.0$ (theory GATE 2). The hidden channel leaves the observable law exactly invariant, so every label-free monitor $M = f(\{\mu\})$ has identical law under fresh and stale deployment: its detection AUC = 0.50 exactly.*

The observable channel tracks regret and is detectable ($AUC = 1.00$); the hidden channel sits at the information-theoretic floor ($AUC = 0.50$). The monitor is therefore not “imperfect” — it is optimal up to a floor no label-free statistic can cross. The only way to see the hidden channel is to read a label: the case for labeled repeatability spot-checks (§8).

6 Applied I — the gap on Fashion-calibrated IVIM posteriors

[PROVISIONAL: this entire section consumes Fashion’s calibrated posteriors.] We apply the gap to a clean synthetic IVIM cohort (Fashion’s simulator, pancreatic-anchor priors; no clinical data) with Fashion-calibrated uncertainty. For the perfusion parameters we compute, on the real (θ, μ, σ) , the coverage scale τ_{stat} , the decision scale τ^* , and the gap, reusing the validated Minos decision core.

The qualitative claim holds. Coverage-calibration is *not* decision-calibration on real Fashion-calibrated posteriors: a gap exists and using the reported bar for the decision leaves substantial regret (up to 3.2 utility units for D^*)^P.

The quantitative law does not transfer — reported, not tuned. The reported σ is itself miscalibrated: mean $\tau_{\text{stat}} = 1.62$ and mean $\text{std}(u) = 1.75$ (the bar *under-disperses* the D^*/f error, reproducing Fashion’s D^* under-coverage and extending it to f)^P. Even after coverage-calibration, the residual decision offset exceeds the leading-order law by a median factor of $17\times$ across the skewed cells^P; signs agree in 10/12 cells, but a $17\times$ magnitude gap is not quantitative support. The cause is honest model mismatch: real IVIM errors are biased ($E[u] \neq 0$), over-dispersed, and heavy-tailed, and the measured skew is generator-dependent — deviations the small-skew, zero-mean, unit-variance idealization abstracts away. The theory is the idealized account; the applied gap is the empirical, miscalibration-dominated reality.

7 Applied II — the validity monitor, observable vs hidden

[PROVISIONAL: reuses Gauge’s deployment monitor and is contextualized by Gauge’s wall.] We apply the label-free monitor (Gauge’s two-family implementation of the Minos-Core idea) to synthetic IVIM with a Gauge-style observable/hidden split (Table 1).

channel	shift	AUC ^P	fires
observable (resolvable)	SNR 40 $\rightarrow$ 12	1.00	yes
observable (resolvable)	tissue- D shift	0.92	yes
observable (resolvable)	perfusion- f shift	0.96	yes
observable (wall-adjacent)	tri-exp perfusion misspec	0.53	yes
hidden	exchangeable; high- D^* conditional failure	0.52	no

Table 1: Observable-fires / hidden-blind on IVIM. The hidden test is exchangeable in the observables, so the monitor is at chance, yet its D^* coverage has collapsed conditionally: marginal 0.89 vs high- D^* tercile 0.79^P.

Shifts that move *resolvable* signal features are detected (AUC 0.92–1.00, all fire), confirming Theorem 2’s observable bound. The exchangeable hidden test sits at $AUC = 0.52$ and does not fire — blind, as Theorem 2(i) predicts — while its high- D^* conditional coverage has failed (0.79 vs nominal). A shift confined to the perfusion regime (tri-exp, AUC 0.53) is only weakly observable, an honest nuance: the blind spot degrades gracefully as drift moves into the unresolvable regime.

8 The observable/hidden throughline

The same split recurs across the program. **Gauge** shows the high- D^* conditional-coverage gap is an irreducible identifiability wall (the Cramér–Rao bound on D^* exceeds the latent-regime width), the empirical instance of Theorem 2’s hidden channel. **Echo** reads sparse labels (repeatability spot-checks) — the only instrument that can cross the label-free floor. Minos supplies the theory that ties them: a cheap label-free monitor is provably optimal up to a floor, and a sparse labeled check is the principled complement, not a redundancy.

9 Discussion and scope

Both theory results are leading-order / idealized (Assumption 1); Theorem 2(i) assumes the hidden channel leaves observables exactly invariant. The applied sections show the *mechanisms* survive contact with real IVIM — a decision-vs-coverage gap, an observable/hidden detectability split — while the clean scaling law is an idealization that biased, over-dispersed IVIM posteriors do not satisfy quantitatively. We report this honestly rather than tune it: the value of the theory is that it names the right objects (a gap; a floor) and predicts their existence and sign, which the applied data confirm.

10 Reproducibility and assumptions

The theory is deterministic and machine-checked (GATE 1–3). The applied sections pin Fashion’s `SEED=0` and Gauge’s `DEFAULT_SEED=20260613`; every number above is regenerated from seeded results by `consistency.py` into `numbers.tex`. The assumptions manifest (`./ASSUMPTIONS.md`) lists every Fashion/Gauge input; one command (`./reproduce.sh`) re-validates the whole build against the final Fashion/Gauge once they publish.

References

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